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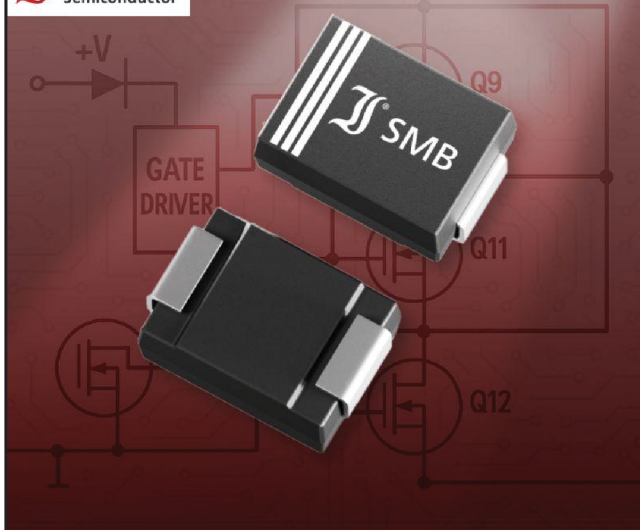
Bootstrap diode with silicon carbide

Bootstrap circuits don't fail dramatically. They fail quietly, and too often. No smoke. Just a high-side MOSFET that refuses to turn on. Sometimes once a day, sometimes once an hour.

Let's zoom in on the scope:

- ▶ A hint of ringing
- ▶ A tiny gate dip
- ▶ UVLO triggers

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Then silence again. It's rarely the driver. Usually, it's the diode no one thinks about: The one that charges the bootstrap capacitor a thousand times a second. At high dv/dt that diode becomes a noise source. A little capacitance here, a little charge there. And suddenly your "stable" design isn't.

We built the SI02C120SMB to end that. A 2A / 1200V SiC Schottky in an SMB package. Low junction capacitance. Almost zero recovery charge. No bootstrap resistor needed. No EMI ghosts in your gate signal. It just works - fast, clean, and repeatable. In power design, stability isn't loud. It's the silence that tells you it's right.

FOR MORE INFO



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